

FLOAT proposal – A shared API between IoT devices and Agroecology Platforms

Pilot Round

Both Proposals (Full and Discovery)

Basics + Summary

- **Project Team**

Mitra Ardron

Dan TerAvest

Greg Austic TBC

Divya Chayanam

- **Collaborating Projects/Organizations**

[Frugal-IoT](#)

[OurSci](#) / [SurveyStack](#)

FarmOS – via OurSci

LiteFarm

(there will probably be non-FLOAT collaborators as well – e.g. other IoT companies)

- **Project Title**

A common API for IoT devices to interact with Agroecology Platforms

- **Problem Statement**

There is high potential, but very little use of IoT sensors in the Agroecology space, because of cost, and lack of integration.

Projects like Frugal IoT and OurSci are building affordable devices for the Agro-Ecology space, that deliver real-time data, or one-off (e.g. lab soil measurement) data.

To be useful data from these devices needs to be available inside agroecology platforms (LiteFarm, FarmOS, etc) where it can be integrated with other data, or used for example for predictions and recommendations.

If, for example, Frugal-IoT just integrates with LiteFarm then we create lock-in for farmers, if they choose to move to FarmOS then their devices become unusable eWaste.

Implementing interfaces between multiple platforms and multiple device projects turns into a big problem quite fast. Each integration requires time and resources by over-stretched and under-funded teams.

- **Project Statement**

The solution is to specify a common interface (aka “API”). Each platform partner can implement one API, and each device partner can implement one API. Then, any farmer can choose their preferred devices, and their

preferred platform, knowing they will work together.

and then any devices can be used with any p

This project would be to bring together as many of the device builders, and the platform builders who want to collaborate.

It would map the needs and aspirations, and also the potential.

It would then figure out a common interface that meets the diversity of needs for both real-time and one-off data.

- **Theme Alignment**

What you can't measure, you can't manage. Integrating device data with farm platforms would expand the range of capabilities of those platforms, without having to resort to using the big commercial ones, or managing device measurement in a separate environment.

- **Alignment with Float's Values and Goals**

I haven't found a written list, but as I understand the Values and Goals:

** Open-Source device projects like Frugal IoT and OurSci bring the ability to measure directly to projects, and out of the hands of Big-Ag.*

** By enabling data to feed into Open-Source platforms it enables farmers and their networks to take control of their own data without having to go through intermediaries who may have different motivations (e.g. to charge you for using your own data, or to sell you something).*

Since both the devices and the platforms are (typically) open-source it brings the potential for control by the users.

With a common API we avoid picking winners, allowing new platforms to arise without the barrier to entry of convincing each device program to interface with them, and vica-versa.

This aligns also with Data and Tech sovereignty as farmers can confidently choose devices or platforms knowing they can in the future switch providers and their devices will still work.

- **Shared Learnings**

The act of collaborating will probably surface additional learnings, both peer-to-peer (one platform learning from another, one device team learning from another) but also a better understanding of the needs of farmers (as understood by platforms) informing device development, and a better understanding of what is affordable and possible informing platform, and farmer's, use of IoT.

- **Future Goals of the Project** *I see the project being high engagement for a relatively brief (e.g. 6 months) followed by a long-term point of coordination (e.g. email lists, and irregular standard revisions) between device and platform teams. The long-term coordination is almost at no-cost, and could potentially be rolled into one of the other FLOAT standardisation projects.*

Float Participation

- **Overall Involvement**

Involvement so far by Mitra Ardrone has been via commenting and interacting on Hylo, and Zoom meetings with other IoT people in the Float community (e.g. Derek; Ian). The time-zones have precluded as much involvement as I would have liked.

Greg & Dan have both been actively involved in several other proposals through their roles with OurSci, GOAT and GOSH.

- **Peer Feedback Involvement**

My (Mitra) conversations with Float members have generally been about how they could use Frugal-IoT to shortcut the development process. AFAIK these have not been submitted as projects. (Ian (Cook Islands) seems to have disappeared, Derrick Ngigi is an advisor). I am not qualified to provide feedback to most of the projects.

- **Project Peer Feedback**

There has been conversation with device-side people in the discussion [topic where this was first proposed](#), that topic no longer seems to be navigable to from the sidebar and its unclear if there is any way to fix that.

Derrick Ngigi (Advisor) gave good feedback.

Each of the relevant parties that have been approached have agreed to participate – that includes OurSci (Dan & Greg) who have existing devices such as their reflectometer, and an aggregator of that data in SurveyStack, and Divya & Hannah at LiteFarm who would look to implement the API to bring sensors into their platform.

I have also had in-principle conversations with one IoT developer in a related field (open-source disaster warning sensors) about the relevance of, for example including their emergency data into farm management tools).

- **Openness to Collaboration**

This project is meaningless without collaboration with other device makers, and with platforms, so it is absolutely looking for collaborators. I (Mitra) do not want to control this, as it would defeat the purpose!. I would suggest adopting the IETF motto “Rough Consensus, and working Code”.

Float Full Proposal Only

Project Details

- **Project Description** (*Describe your project in detail (up to 400 words).*)

The overall goal is an interoperable set of APIs, Standards, Protocols, procedures etc that enable any relevant device producer – such as Frugal IoT or OurSci to have their real-time and/or one-off (lab) data integrated into any relevant platform such as LiteFarm

The project would:

Stage 0: In very Short Term – i.e. pre funding.

- Bring together device builders (Frugal IoT, OurSci and others involved, or outside of FLOAT) with platform providers (LiteFarm and others involved or outside of FLOAT).

Stage 1: Needs assessment.

- Do a simple needs assessment for integration – mostly how platforms want to use data, but also any technical constraints.
- Deep dive into those needs – which could go into topics like: Security; Privacy; Data integrity; etc

- The assumption is that needs will vary, but that there is a commonality of needs and capabilities that can realistically be met.
- Success will be a common understanding of meetable needs, and also what cannot be done in the first round of the API/standard.

Stage 2: Development of the standard

- Develop a common API/protocol and iterate until it meets as many of the needs as possible for each of the participants,
- Ensure it is as compatible as possible with other collaboration standards, ontologies etc – (Common Farm Convention has been mentioned but not yet evaluated)
- This is the high-effort part of the project, calls, drafts, working through problems, multiple iterations.
- Success would be a draft everyone is at least 90% happy with.

Stage 3: Implementation

- Each partner would then commit to implement their end of the API.
- There will almost certainly be further rounds of iteration, as a theoretical API rarely holds up precisely when implemented.
- Success would be at least one functioning integration between devices and platform that can be trialed in the field.

Stage 4: Documentation

- Update the API to integrate learnings from Stage 3.
- Document the API and how to work with the different partners.
- Success would be a document & process that a new participant could follow to integrate either their platform, or their device.

Medium term – i.e. after June 2026, and so dependent on other funding

- Publicise to non-participants, for example this could be device makers – in Western, or less affluent contexts.
- Facilitate those new participants to join the process.
- Success would be a new participating device developer or platform integrated into all relevant other parties.

Long Term

- In the long term this becomes a low-effort open-source standards committee.
- Maintain an online space (e.g. Github repo) where documents can be maintained
- Produces new versions as needs change.
- If something broader evolves (such as the GOAT proposal) it could easily become a working group of that.

• **Measurable Outputs**

(Describe the measurable outputs, if any, which could serve as a reasonable proxy for project success, or describe why this is difficult for this project specifically (up to 100 words))

- Needs assessment of the various participants

- Draft API/Standard/Protocol with rough consensus that is sufficiently detailed for the various participants to start implementing
 - Working integrations between at least one set of devices, and one platform.
 - N * N integration – i.e. all participating device teams integrated with any relevant participating platforms.
 - Final V1 of API/Standard/Protocol reflecting learnings of the development phase.
 - Documentation (for new participants) on how to work with the different partners.
- **Anticipated Challenges** *Share any anticipated challenges that could keep the project from accomplishing aims, and how you plan to mitigate them (up to 200 words).*
- Getting engagement from relevant parties:
 - Reach out via FLOAT and help each see the advantage of collaboration.
 - Time, availability and priority of the participants
 - Unless the participants see this as a priority, and allocate management and developer time to it, the project will founder
 - If at least one device-side participant and one platform-side participant are committing resources it can move forward.
 - Consensus of needs – projects are all doing different things
 - Conversations – working through how to meet most of the needs of all of the participants. I think this is helped by the generally collaborative nature, and non proprietariness of this community.
 - Contradictor technical needs – e.g. one team uses MQTT, another uses HTTP/REST or a need to use different identity providers.
 - Put the engineers on it ! Most of these problems have workable solutions when there is good will.
 - Compromises
 - This can be tougher – e.g. meeting a higher level of security needed by one participant could be a problem for another participant whose primary value might be affordability to low income smallholders.
- **Team Expertise** *Describe your expertise, extra special gifts, and affiliations of the Project Team (up to 200 words).*

Mitra Ardron: Has been actively involved in developing standards, APIs and protocols since the early days of the internet including being an active contributor (HTTP); lead author (VRML); Standard committee chair (URL) etc. Mitra now leads the open-source Frugal-IoT project, an 18 month old project to develop affordable sensors for resource constrained environments, including, but not limited to regenerative agriculture and rural development. Mitra founded a company that built embedded payment technology for small solar systems used mostly across Africa and has long involvement

in other affordable technologies for the SDGs.

Dan TerAvest: A soil scientist with experience bridging the divide between tool development and user experience, coordinating between technical and user teams to develop appropriate tools and methods. Dan has actively supported development and deployment of sensors and decision support tools in smallholder settings in Africa and Central America.

Greg Austic: An open source technology advocate who is a co-organizer of the GOSH and GOAT communities. He played a key role in development of the Our Sci Reflectometer and collaborated with Purdue University, USDA and OpenTEAM to develop the common farm convention, a data schema for farm management data.

Divya Chayanam: Product Lead at LiteFarm, an open-source farm software. A 0-1 product manager specializing in launching and scaling early stage digital products.

Project Impacts

- **Anticipated Impacts** *Describe the project's anticipated impacts across interdependent scales (individuals, farmers, and users; communities and territories; ecosystems; economies – up to 400 words).*

As an enabling technology impacts are by necessity indirect.

Better and Affordable Device Choices – If successful then projects using, for example, Frugal-IoT devices, will have a choice of different platforms to work with the data in, and ways to integrate with other farm management functionality. Similarly users of, for example LiteFarm, would be able to choose to use (and mix) devices from any participating device project.

Interoperability – The API ensures that a device data can freely integrate across different software.

- **Key Beneficiaries** *Who specifically benefits, and how? (up to 200 words)*

The most obvious beneficiaries are farmers, who could choose their preferred devices, and independent choose their preferred platform, knowing the two will work together, and that they won't be locked in to a particular supplier.

A secondary beneficiary is the environment. Lock-in means that a potential switch in platform leads to devices being unusable eWaste unable to work on the new platform.

Again, this is hard to to specify. The most obvious specific (example) beneficiary would be a project we are working with to support regenerative/organic transition for smallholder farmers. Instead of being locked into Frugal-IoT's back-end, they could choose to integrate with LiteFarm, or any other participating platform.

- **Unintended Consequences** *What if you are wrong? What are potential consequences – downsides, risks, or possible harms that could result from implementing or attempting to implement this project? (up to 200 words)*

It's unclear of unintended consequences. If we over-complicated the protocol, we could push the cost of devices up; if it was made too rigid it could stifle innovation.

Project Plan

- **Key Milestones and Deliverables** *Provide a clear timeline for your project with key milestones and deliverables (up to 200 words).*

The stages are outlined above – along with deliverables. My estimated timeline would be ...

Stage 0 – reaching out pre funding
 Stage 1 – Needs assessment: January
 Stage 2 – Standard development: February – March
 Stage 3 – Implementation: April – May
 Stage 4 – Documentation: June
 Medium term – Outreach & new partners: July – Dec
 Long term – maintenance 2027...

- **Budget** *Provide a detailed budget for your project.*

This is an estimate of the time required –

Task	Amount	Person-Hours
Process coordination	1hr/week * 35 weeks	35 hours
Standard drafting, editing and compilation	12 hours in Stage 1 36 hours in Stage 2 12 hours in Stage 3	60
Documentation	25 hours in Stage 4	25
Implementation	4 projects * 45 hours	180
Total		300 person-hours
Rate	At US\$65/hr	US\$19,500

- **Budget Narrative** *Explain how the requested funds will be used to achieve your objectives (up to 100 words).*

The requested funds would be split across the work as detailed above and below, this could be one person for each key task, or multiple people sharing the work.

Standards work largely progresses through volunteer involvement by the different parties. However, they tend to flounder, or get delayed, unless

there is sufficient coordination – e.g. setting up meetings, circulating minutes, chasing people for commitments etc. I am guessing this takes about 1hr/week for a period slightly longer than the project.

The biggest job is editing all the different ideas into the common proposal, this includes:

- * in Stage 1: compiling the needs assessment and drawing out commonality;
- * in Stage 2 editing a protocol or API document;
- * in Stage 3 revising that document as implementation surfaces problems;

In order to ensure other new participants can join in later, documentation needs producing, and at the very least there is either writing, or editing submissions from various volunteers.

The implementation stage would ideally be 4 projects – two on the device side, two on the platform side, each receiving funding for 45 hours development, which is probably a reasonable estimate for integrating with internal systems, and coordinating with the other partners.

If the full amount is not funded, then there may be opportunity for the participating organizations to fund some of their own participation in the development phase, however since all the organizations are themselves non-profits this may then have to compete with other priorities, adding a delay.

- **Additional Resourcing** *To help inform future Float funding rounds, beyond funding, what other resources or support would be helpful for your project's success? (e.g., technical expertise, community feedback, mentorship) (up to 100 words)*

The most useful thing to this project would be the participation of other relevant parties, especially during the Needs Assessment phase.

Discovery Round

- **Participation in discovery round:** If your proposal is not selected, would you like it to be automatically included in the “Discovery” co-budgeting round, which is set to follow?

YES